

Unit 04



Main tasks in Scientific Research



1. ACADEMIC READING

ACADEMIC READING



- Reading for academic and research purposes is very different from reading for leisure.
- An academic book isn't written in a way that it needs to be read from page one to the last page. It's written to allow you to dip in and find the information you need.
- It is important that you develop and improve your skills for reading academically, so that you can **pick out relevant information more effectively**.

PREPARING TO READ



- Before reading, ask yourself the following questions:
 - Why am I reading this?
 - What do I want to find?
 - What information do I already know, and will the text 'fill in the gaps' for me?
- Remember books, chapters and articles have not been written especially so that you can answer your question. They may go into a lot of complicated depth which is not relevant to your current needs.
- Consider the date when a book or article was written and ask yourself if this is the latest information and research, or is it now out-of-date. Books published many years ago can still be extremely valuable for certain subjects, but will reflect the scholarship of their age.

STYLES OF READING



- You need to vary your reading style and speed according to the material you are reading.
- There are three styles of reading, which we use in different situations:
 - scanning,
 - skimming
 - and detailed reading.



- Scanning is most useful when you are **searching for something specific** in the text – like a word or phrase. An example of scanning is when you look in the index of a book to see which page contains the information you want.
- You can use the same skill in academic reading to find the material that is relevant to you.
 - Move your eye quickly over the page to find particular words or phrases that are of significance to the task you are doing.
 - This will provide you with an overview of the reading matter so you can decide if it is relevant to your research.
 - Always ask yourself if the material will be helpful for your assignment.



- It is useful to scan key parts of texts to see if the material will be useful to you:
 - The introduction or preface of a book
 - The first line of each new paragraph in a chapter or article
 - The concluding chapter of a book or paragraph of an article
 - The abstract of a journal, which contains a summary of the article and the rationale for the study

SKIMMING



- Skimming is a visual activity and is used for **getting the gist** or impression of a chapter or section of text in the shortest time possible.
- Aim to read quickly to get the main points and skip over the detail.
- It is useful to skim:
 - To preview a passage before you read it in detail
 - To refresh your understanding of a passage after you have read it in detail
 - To decide if a book in the library or bookshop is right for you

DETAILED READING



- Detailed reading is used **for extracting information accurately**. This is where you read text more closely, and work to learn from the text.
- You will do careful reading after you have scanned the text to see if it is relevant and skimmed the text to get a general idea of the topic.
- When reading in detail:
 - Ask yourself questions and search for the answers in order to focus your reading
 - Read the article critically and analyze and evaluate the findings



- When reading in detail:
 - Avoid 'sounding out' words in your head as you read. This slows you down.
 - Avoid **backtracking** when reading.
 - Backtracking is when you read a few words and then go back over them because you have not understood the point properly or were not concentrating.
 - By doing this you are interrupting the flow of reading.
 - It is far better to get to the end of a section by reading it straight through and then re-reading it if necessary.

TIPS for ACTIVE READING



- **Underlining and highlighting**

As you are reading, you will have to make decisions about the information in order to select what is most important (if you have highlighted an entire page then you are not being selective or critical enough). Highlight text after you have read a passage to make sure you are only highlighting the key information.

- **Note key words or points**

Record the main headings and subheadings as you read. Use one or two keywords for each point. If you are using photocopies, you can write a key word beside each paragraph so you can scan later for the relevant section.

- **Questions**

Before you start reading an article, a chapter or a whole book, prepare for your reading by noting down questions you want the material to answer.

TIPS for ACTIVE READING



- **Summaries**

Pause after you have read a section of text. Then:

- put what you have read into your own words
- skim through the text and check the accuracy of your summary
- fill in any gaps

- **Post-it notes**

Use Post-it notes to mark pages and quotes of interest to which you may like to return.

- You should learn a huge amount from your reading. You will get far more benefit from it if you train your mind to learn rather than if you read passively.

STRUCTURE OF TEXT

- Finding the answers to the following points will mean that you have to examine the way text is put together, and this will help with comprehension and speed of access to information:
 - Where are chapter summaries usually found – at the beginning or the end of a chapter?
 - Do the sub-headings give me an overview of the structure of the chapter?
 - How is key terminology presented – bold, underlined or a separate glossary?
 - Does each section contain a summary statement at the beginning or the end?
 - Does each section have subsidiary and supporting material or evidence or examples after summary statements?
 - Are diagrammatic features, tables and graphs used to explain prose text?
 - Is the sequencing of the information obvious in the text layout?

CRITICAL READING



- To read critically is to make judgements about **how** a text is argued.
- The key to critical reading:
 - Do not read only for information.
 - Do read for ways of thinking about the subject matter.
- You might have to read a text through once to get a basic grasp of content before you launch into an intensive critical reading.
- Whatever you are reading, you should find out and be aware of the author's background.

CRITICAL READING



- Try this for yourself. Choose a chapter or an article and find out answers to the following questions:
 - Who is the author's audience? What are the central claims of the text? What is the main evidence?
 - How has the author analyzed the material to set up an argument? How is this substantiated?
 - What assumptions lie behind the evidence or arguments? Do you think the assumptions are founded on adequate proof?
 - What methodology was used? What are the general weaknesses or strengths?
 - What do other leading thinkers or writers have to say about this?



- **What is your evaluation?**
 - You need to consider and decide if the arguments and accompanying evidence are strong or weak
 - Do you think the argument could have been made differently or supported in a different way?
 - Can you spot any gaps, un-argued assumptions or inconsistencies?
 - Look at the conclusions and ask yourself if the evidence supports the conclusions



2. RESEARCH METHODOLOGIES

RESEARCH METHODOLOGIES in COMPUTING SCIENCE

- Computing Science researchers use several methodologies to tackle questions within the discipline.
- Several of these methodologies are:
 - Formal
 - Experimental
 - Build
 - Process
 - Model
- Tasks performed by a single researcher fall within different methodologies. Even the activities required to tackle a single research question may include several of these methodologies.

FORMAL METHODOLOGY



- In Computing Science, formal methodologies are mostly used to prove facts about algorithms and system.
- Researchers may be interested on the formal specification of a software component in order to allow the automatic verification of an implementation of that component.
- Alternatively, researchers may be interested on the time or space complexity of an algorithm, or on the correctness or the quality of the solutions generated by the algorithm.

EXPERIMENTAL METHODOLOGY



- Experimental methodologies are broadly used in CS to evaluate new solutions for problems.
- Experimental evaluation is often divided into two phases.
 - In an exploratory phase the researcher is taking measurements that will help identify what are the questions that should be asked about the system under evaluation.
 - Then an evaluation phase will attempt to answer these questions.
- A well-designed experiment will start with a list of the questions that the experiment is expected to answer.

BUILD METHODOLOGY



- A “build” research methodology consists of building an artifact - either a physical artifact or a software system - to demonstrate that it is possible.
- To be considered research, the construction of the artifact must be new, or it must include new features that have not been demonstrated before in other artifacts.

PROCESS METHODOLOGY



- A process methodology is used to understand the processes used to accomplish tasks in Computing Science.
- This methodology is mostly used in the areas of Software Engineering and Man-Machine Interface which deal with the way humans build and use computer systems.
- The study of processes may also be used to understand cognition in the field of Artificial Intelligence.

MODEL METHODOLOGY



- The model methodology is centered on defining an abstract model for a real system.
- This model will be much less complex than the system that it models, and therefore will allow the researcher to better understand the system and to use the model to perform experiments that could not be performed in the system itself because of cost or accessibility.
- The model methodology is often used in combination with the other four methodologies.
- Experiments based on a model are called simulations.
- When a formal description of the model is created to verify the functionality or correctness of a system, the task is called model checking.



3. WRITING A RESEARCH PROPOSAL

RESEARCH PROPOSAL



- A research proposal is a summary of what you want to do, why and how.
- Before starting a research program it is helpful to make it clear to yourself:
 - what the aims of the research are
 - and what you need to do to achieve the desired outcome.
- A research proposal will also be useful as a way to inform others of your intentions.

RESEARCH PROPOSAL



- A research proposal needs to explain the nature of the research and its context, and why it is needed.
- This then prepares the way for:
 - a statement of the aims,
 - objectives of the research,
 - how it will be carried out,
 - and what the outcomes are likely to be.
- Then there is a description of what resources will be required (time, money, equipment, etc.) in order to carry out the work.

RESEARCH PROPOSAL



- Academic research proposals are usually composed of the following elements:
 - the title;
 - aims of the research;
 - the background to the research – context and previous research;
 - a definition of the research problem;
 - outline of methods of data collection and analysis;
 - possible outcomes;
 - timetable of the project and description of any resources required;
 - list of references.



- The function of the title is to encapsulate in a few words the essence of the research.
- Ideally it should contain all the essential key words that someone might use in an attempt to locate the kind of study you are proposing.
- These words are likely to include the main concept and variables, and limits to the scope.
- You can leave out such phrases as ‘an investigation into’, ‘a study of’, ‘aspects of’, as these are obvious attributes of a research project.
- Example: “First-year undergraduate calculus students: Understanding their difficulties with modeling with differential equations.”

AIMS OF THE RESEARCH

- One main aim, and perhaps two or three subsidiary aims resulting from it are sufficient.
- Precision is essential – too many or vague aims indicate woolly thinking and will be difficult or impossible to achieve within the available resources and time.
- Example:
Specifically, the proposed research is based on the notion that since “[m]any students struggle over the same hurdles in the same sequence in learning the same material ... descriptive analyses of conceptual understanding are not only feasible, but likely to be widely applicable” (Trowbridge & McDermott, 1980). This notion is reinforced by the successes of this approach in the reform efforts in the closely related field of physics education (see McDermott & Redish (1999) for a review). Consequently, the main aim of this research is to identify and describe the major conceptual difficulties mathematics students have understanding mathematical models of physical problems which involve the use of first-order ordinary differential equations.

THE CONTEXT – BACKGROUND AND PREVIOUS RESEARCH

- This explains to the reader the background from which the research problem emerges.
- It should explain the major factors which surround your problem, and of any significant literature which relates to it.
- Virtually every subject has been researched previously, so a critical account of what has been achieved so far to address the problem is required in order to identify the gaps in knowledge or contentious issues.
- The research should fill one of these gaps or try to resolve the contention.

THE RESEARCH PROBLEM



- The research problem provides the focus of the research project.
- It is the culmination of the background work and the initiator of the specific research tasks.
- It must be very clearly defined to explain the nature of the problem and why it is significant.
- The problem may be expressed in abstract **terms** initially, but through the statement of **sub-problems**, you should indicate how it can be investigated practically.

THE RESEARCH PROBLEM

- Example:

Title: Task-Oriented Pattern Discovery for Predictive Web User Modeling

Research Problem: An essential task in building personalized and adaptive systems is the automatic discovery of predictive models for user behavior. Existing approaches, such as clustering, correlation analysis, and association discovery, tend to generate shallow patterns which do not capture the full complexity of users' online behavior. Nor can the generated patterns explain the users' underlying interests which lead to specific types of behavioral patterns. To better capture users' underlying interests or information needs, we introduce the notion of “task”. A task is a set or sequence of actions which are likely to be performed commonly by users in order to meet a specific information need or perform a specific function. These tasks are not directly visible, but can be captured and characterized either by a combination of users' interactions with the site and the site's content and structure. (cont.)

THE RESEARCH PROBLEM

- Example:

Title: Task-Oriented Pattern Discovery for Predictive Web User Modeling

Research Problem: (cont.) We hypothesize that patterns discovered at the task level can provide a better understanding of users' underlying interests, which in turn, can lead to better predictive models. We propose a framework for “Task-Oriented Web User Modeling”. We intend to use probabilistic latent variable modeling to automatically discover and quantify user “tasks” and task-level patterns from users’ navigation data, as well as from Web site's content and structure data. Based on this framework, we will propose a novel personalization approach, based on the maximum entropy principle, which allows for a seamless integration of content-based and usage-based task-level patterns. We will perform experiments on real Web usage data and movie rating data to validate that the proposed approach results in more accurate and flexible predictive models.

OUTLINE OF METHODS



- This part of the proposal explains briefly what you are going to do in order to carry out your research, based on your chosen research approach.
- Every proposal is different in its description of methods, as these have to be specifically tailored to efficiently collect and analyze data relevant to the specific research problem and to produce the outcomes aimed at.
- The explanation will outline which methods you will use for what, and why.
- There may be only one method used, e.g. experimental, or a combination of several, each suitable for different aspects of the research.

POSSIBLE OUTCOMES OF THE RESEARCH

- You should try to be quite precise as to the nature and scope of the outcomes and as to who might benefit from the information.
- The outcomes should relate directly to the aims of the research.
- Note that in PhD proposals and most funded research there is a need to indicate what will be the original contribution to knowledge.

POSSIBLE OUTCOMES OF THE RESEARCH

- Example:

Title: : “First-year undergraduate calculus students: Understanding their difficulties with modeling with differential equations.”

Possible Outcomes: The expected outcome is to be able to describe in detail the most commonly occurring conceptual difficulties students have with modeling with differential equations for a wide range of modeling situations. These outcomes will help instructors in this area develop learning activities which will help students overcome these conceptual difficulties, and thereby not only improve the learning experience of students, but also increase the chances that reform efforts using modeling with ODEs will prove fruitful rather than another disappointing fad. In particular, if the research supports the various hypotheses, various exercises which draw students' attention to the discriminations they need to make are fairly evident and could be tested in future research as to how effective they are in helping students make the appropriate paradigm shifts.

TIMETABLE OF THE PROJECT AND DESCRIPTION OF ANY RESOURCES REQUIRED

- There is always a time limit to research projects. The formulation of a timetable is therefore essential in order to allocate time limits to the sequence of tasks that you need to accomplish in order to hand the finished report or dissertation in on time.
- Resources that are essential to the project, such as equipment, transport, skills, software etc. should be listed and availability assured in order to convince the reader (and yourself) of the practicality of achieving the aims of the project.

LIST OF REFERENCES



- The location of the cited work of others must be meticulously recorded in this list.
- Not only does this ensure that you cannot be accused of plagiarism but also demonstrates that you are aware of the latest thinking in the subject.
- However, keep the references relevant – it is not a bibliographic list of all the literature surrounding the subject.

WHY IS A PROPOSAL REJECTED?



- The scale of the proposed research is too big so an adequate conclusion cannot be reached.
- The aims and objectives are vague or impractical or do not match or are expressed as personal goals.
- The research problem is not clearly defined so the research work has no adequate focus.
- There is a lack of logic in the argument in the proposal, so the link between the research aims and outcomes is broken.

WHY IS A PROPOSAL REJECTED?



- Access to information, equipment or other resources are uncertain or impossible.
- Not enough information is provided about the details of the project.
- The project is formulated on an emotional or a political agenda rather than a factual or a theoretical basis.
- The procedures are confused with the objectives.



4. THE NATURE OF DATA

THE NATURE OF DATA



- Data is another word for **bits of information** (singular – datum).
- Research uses data as the raw material in order to come to conclusions about some issue.
- All research involves the collection and analysis of data, whether through reading, observation, measurement, asking questions, or a combination of these or other strategies.
- It depends on the issue being investigated what data needs to be collected. These data may vary considerably in their characteristics.

THE NATURE OF DATA

- Data are not only **elusive**, but also **ephemeral**.
 - They may be true for a time in a particular place as observed by a particular person, but might be quite different the next day.
 - Take for example, a daily survey of people's voting intentions in a forthcoming general election. The results will be different every day even if exactly the same people are asked, because some change their minds because of what they have heard or seen in the interim period.

THE NATURE OF DATA

- Data are not only ephemeral, but also **corruptible**.
 - Hearsay, second hand reports and biased views are often paraded as facts.
 - The further away you get from the event the more likely it is that inconsistencies and inaccuracies creep in.
 - Memory fades, details are lost, recording methods do not allow a full picture to the given, and distortions of interpretations occur.

LEVELS OF ABSTRACTION



- Data are part of a hierarchy of information, going from the general to the particular, from abstract to concrete.
- Understanding this hierarchy makes it possible to break down research problems expressed in theoretical language to more practical components that can be measured in some way.

LEVELS OF ABSTRACTION



- This hierarchy can be expressed like this:
 - **Theory** – abstract statements that make claims about the world and how it works. Research problems are usually stated at a theoretical level.
 - **Concepts** – building blocks of the theory which are usually abstract and cannot be directly measured.
 - **Indicators** – phenomena which point to the existence of the concepts.
 - **Variables** – components of the indicators which can be measured.
 - **Values** – actual units of measurement of the variables. These are data in their most concrete form.

LEVELS OF ABSTRACTION

- The briefest statement of a theory will be the most general and abstract, whilst the detailed components of the statement will become increasingly particular and concrete.
- For example:
 - **Theory** – poverty leads to poor health.
 - **Concepts** – poverty, poor health.
 - **Indicators of poverty** – low income, poor living conditions, restricted diet, etc.
 - **Variables of poor living conditions** – levels of overcrowding, provision of sanitary facilities, infestations of vermin, levels of litter, etc.
 - **Values of levels of overcrowding** – numbers of people per room, floor areas of dwellings, numbers of dwellings per hectare, etc.

LEVELS OF ABSTRACTION



- Data, seen as bits of information, can be at any level of abstraction.
- Research projects usually start at the more abstract end of the spectrum, move to the more concrete during the investigation, and return to the abstract in the conclusions.
- Data that can be manipulated, measured and analyzed tends to be more at the values level.

LEVELS OF ABSTRACTION

You can relate these levels of abstraction to how to structure your research.

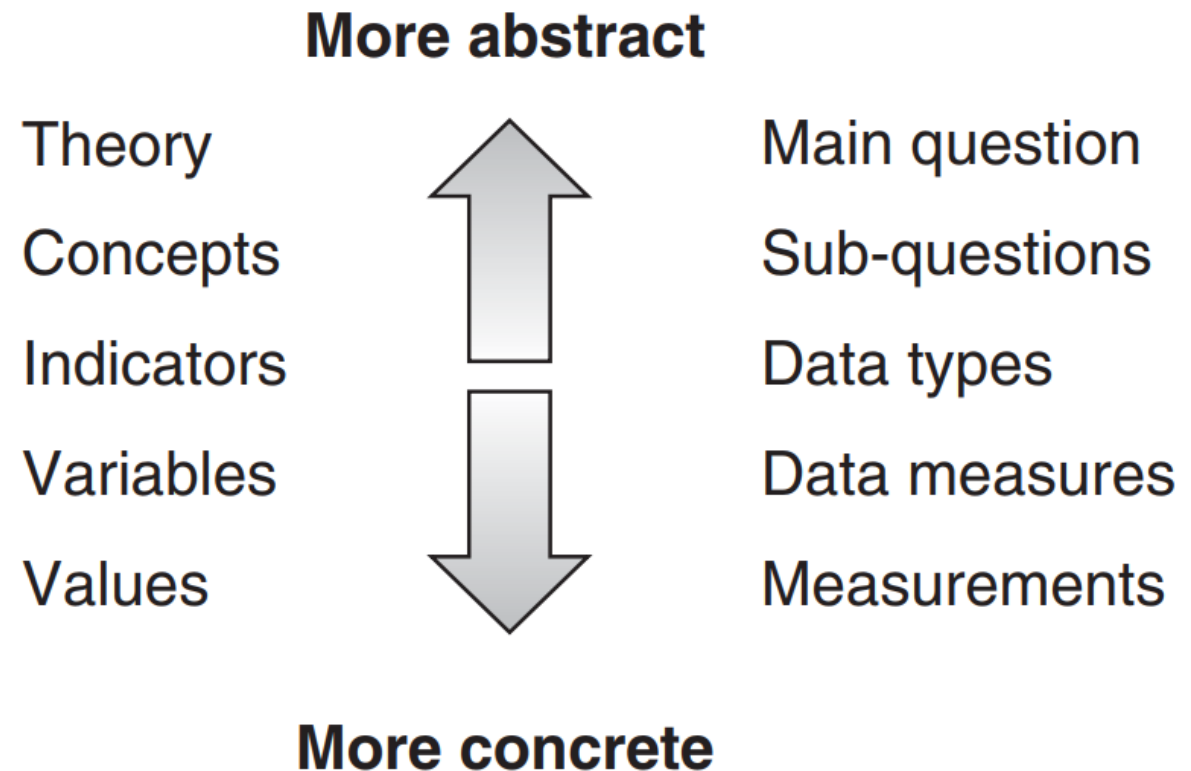


Diagram of levels of abstraction

PRIMARY AND SECONDARY DATA

- Data come in two main forms, depending on its closeness to the event recorded.
- Data that has been observed, experienced or recorded close to the event are the nearest one can get to the truth, and are called **primary data**.
- Written sources that interpret or record primary data are called **secondary sources**, which tend to be less reliable.
- For example, reading about a fire in your own house in the newspaper a day after will probably give you less accurate information than what you gained by experiencing the event yourself. You will be more informed about the facts and these will not be distorted by someone else's interpretation.

- There are four basic types of primary data, distinguished by the way they are collected:
 - **Measurement** – collections of numbers indicating amounts, e.g. voting polls, exam results, car mileages, oven temperatures etc.
 - **Observation** – records of events, situations or things experienced with your own senses and perhaps with the help of an instrument, e.g. camera, tape recorder, microscope, etc.
 - **Interrogation** – data gained by asking and probing, e.g. information about people's convictions, likes and dislikes etc.
 - **Participation** – data gained by experiences of doing things e.g. the experience of learning to ride a bike tells you different things about balance, dealing with traffic etc., rather than just observing.

PRIMARY DATA



- The primary data are the first and most immediate recording of a situation.
- However, collecting primary data is time consuming and not always possible.
- Although more data usually means more reliability, it is costly to organize large surveys and other studies.
- Furthermore, it is not always possible to get direct access to the subject of research. For example, many historical events have left no direct evidence.

SECONDARY DATA

- Secondary data are data that have been interpreted and recorded.
- The quality of the data depends on the source and the methods of presentation.
- It is quite common in student level research to rely on secondary data for the actual research investigations rather than generating new primary data from the field.
- Wherever there exists a body of recorded information, there are subjects for study.
- You can imagine using existing resources when doing an historical study (i.e. of any past events, ideas or objects, even the very recent past) or a nationwide or even a local study that uses official statistics as the principle data.

SECONDARY DATA



- The advantage of using sets of secondary data is that it has been produced by teams of expert researchers, often with large budgets and extensive resources way beyond the means of a single student, so it cuts out the need for time consuming fieldwork.
- Data that has been collected over a long period of time will provide the opportunity to do a longitudinal study (tracing the developments over time), impossible to do with data collected in short projects.
- Data in the public realm are also open to scrutiny by others and is a permanent resource on which to base your research.

SECONDARY DATA

- There are several online sites that provide access to data sets from a variety of sources:
 - **Google's Datasets Search Engine** (<https://toolbox.google.com/datasetsearch>)
 - **Microsoft Research Open Data** (<https://msropendata.com/>)
 - **The U.S. Government's open data** (<https://www.data.gov/>)
 - **UCI Machine Learning Repository** (<https://archive.ics.uci.edu/ml/index.php>)
 - **Open Data on AWS** (<https://registry.opendata.aws/>)
 - **Kaggle Datasets** (<https://www.kaggle.com/datasets>)
 - ...

QUANTITATIVE AND QUALITATIVE DATA

- Data are also divided into two other categories, referring not to their source but to their characteristics; basically whether they can be reduced to numbers or presented only in words.
- Numbers are used to record much information about science and society, for example pressures, bending forces, population densities, cost indices etc. This type of data is called **quantitative data**.
- People's judgements, feelings of comfort, emotions, ideas, beliefs etc. can only be described in words. These record qualities rather than quantities, hence they are called **qualitative data**.

QUANTITATIVE DATA

- Quantitative data can be measured, more or less accurately because it contains some form of magnitude, usually expressed in numbers.
- You can use mathematical procedures to analyze the numerical data.
- These can be extremely simple, such as counts or percentages, or more sophisticated, such as statistical tests or mathematical models.
- Census figures (population, income, living density, etc.), economic data (share prices, gross national product, tax regimes, etc.), performance data (e.g. sport statistics, medical measurements, engineering calculations, etc.) and all measurements in scientific endeavor are all typical examples of quantitative data.

QUALITATIVE DATA



- Qualitative data cannot be accurately measured and counted, and are generally expressed in words rather than numbers.
- Qualitative research depends on careful definition of the meaning of words, the development of concepts and variables, and the plotting of interrelationships between these.
- Observation notes, interview transcripts, literary texts, minutes of meetings, historical records, memos and recollections, documentary films, are all typical examples of qualitative data.
- Research, particularly when about human beings, often combines the examination of both qualitative and quantitative data.

MEASUREMENT OF DATA



- Data can be measured in different ways depending on their nature.
- These are commonly referred to as levels of measurement – **nominal**, **ordinal**, **interval** and **ratio**.
- You can use the following simple test to determine which kind of data measurement that you can use on the values of a variable. If you can say that:
 - one value is different from another, you have a **nominal scale**;
 - one value is bigger, better or more of anything than another, you have an **ordinal scale**;
 - one value is so many units (degrees, inches) more or less than another, you have an **interval scale**;
 - one value is so many times as big or bright or tall or heavy as another, you have a **ratio scale**.

NOMINAL LEVEL



- Nominal measurement is very basic – it divides the data into separate categories that can then be compared with each other.
- For example, buildings may be classified into many types, e.g. commercial, industrial, educational, religious etc.
- What is important is that every category is distinctive, that there is no overlap between them which makes it difficult to decide where to place a particular piece of datum.
- Nominal data can be analyzed using only simple graphic and statistical techniques. Bar graphs, for example, can be used to compare the sizes of categories and simple statistical properties such as the percentage relationship of one subgroup to another or of one subgroup to the total group can be explored.

ORDINAL LEVEL



- This type of measurement puts the data into order with regard to a particular property that they all share, such as size, income, strength, etc.
- Precise measurement of the property is not required, only the perception of whether one is more or less than the other.
- For example, the runners in a marathon can be sorted by the order in which they finished the race; or we can measure members of the workforce on an ordinal scale by calling them unskilled, semi-skilled or skilled.
- The ordinal scale of measurement increases the range of statistical techniques that can be applied to the data.

INTERVAL LEVEL



- With this form of measurement, the data must be able to be measured precisely on a regular scale of some sort, without there being a meaningful zero.
- For example temperature scales, in the Fahrenheit, Celsius and Rankine scales, the gradation between each degree is equal to all the others, but the zero point has been established arbitrarily. They each precisely measure the temperature, but the naught degrees of each are different.
- The interval level of measurement allows yet more sophisticated statistical analysis to be carried out.

RATIO LEVEL

- The ratio level of measurement is the most complete level of measurement, having a true zero: the point where the value is truly equal to naught.
- Most familiar concepts in physical science are both theoretically and operationally conceptualized at a ratio level of quantification, e.g. time, distance, velocity, mass etc.
- A characteristic difference between the ratio scale and all other scales is that the ratio scale can express values in terms of multiples of fractional parts, and the ratios are true ratios. For example, a meter is a multiple (by 100) of a centimeter distance, a millimeter is a tenth (a fractional part) of a centimeter. The ratios are 1:100 and 1:10.
- Of all levels of measurement, the ratio scale is amenable to the greatest range of statistical tests.

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